

Agentic Retrieval-Augmented Generation: A Comparative Survey

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Abstract—Retrieval-Augmented Generation (RAG) has proven to be an effective method for enhancing the factual accuracy of large language models (LLMs) by incorporating external knowledge into the response generation process. Traditional RAG methods are typically based on static, one-step retrieval. In contrast, Agentic RAG introduces iterative retrieval, planning, memory, and autonomy mechanisms to tackle more difficult queries and improve the system’s ability to operate within the dynamic context of the information retrieval process itself. In this paper, we perform a comparative analysis of the most common Agentic RAG approaches, namely, goal-oriented, multi-agent, graph-based, and adaptive methods. We discuss their benefits in terms of improved reasoning and reduced hallucination but also analyze their limitations in terms of complexity, additional costs, and scalability. Finally, we present a reinforcement learning-based approach to adaptive retrieval-augmented generation that utilizes a hierarchical policy to learn policies for retrieval strategy selection at different levels of abstraction, enabling more effective decision-making and improved adaptability. Our aim is to provide a comprehensive overview of Agentic RAG approaches and identify research directions that hold promise for future work. In summary, this paper aims to provide an understanding of Agentic RAG approaches and their potential benefits and limitations. By comparing approaches and exploring a novel reinforcement learning-based solution, the paper contributes to the growing body of research on improved methods for factual accuracy in LLMs.

Index Terms—Agentic RAG, Retrieval-Augmented Generation, Large Language Models (LLMs), Reinforcement Learning, Multi-Agent Systems, Adaptive Retrieval, Hallucination Mitigation

I. INTRODUCTION

Recent advances in Large Language Models (LLMs) have led to impressive gains in natural language understanding and generation. Nevertheless, these models frequently produce factually incorrect or unsupported information, especially when operating outside their training distribution. This limitation is particularly problematic in settings that require a high degree of responsiveness and reliability, such as healthcare and finance. The recently proposed approach of Retrieval-Augmented Generation (RAG) offers a promising solution to this challenge, by combining elements of information retrieval and natural language generation [1], [2]. Following up on the initial works, multiple recent surveys recognize the essential role of retrieval-augmented approaches in large language models for enhancing factual accuracy and external knowledge grounding [3], [4]. By design, RAG-based methods provide a pathway towards more reliable and factual responses

by grounding language generation in external knowledge. However, the majority of RAG approaches are essentially static in nature, only performing a one-step retrieval before generation. These methods cannot effectively reason about complex questions that require multi-step logical or conceptual dependencies to be resolved and thus fail to adaptively update the retrieved context. To fulfill the need for more expressive and responsive question-answering systems, we therefore propose to enhance Agentic RAG with reinforcement learning.

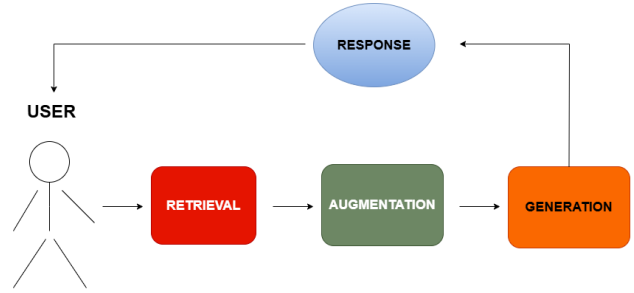


Fig. 1: Basic Components of RAG

Agentic Retrieval-Augmented Generation (Agentic RAG) is one recent approach that attempts to address these challenges [5]. Recent reasoning and tool-augmented methods such as ReAct, Toolformer, and WebGPT utilize language models’ capacities to reason and use external tools or retrieve information from databases to achieve better performance, while also grounding their responses in the real world [6]–[8]. By leveraging reasoning, memory, and tools, recent models are capable of performing tasks that require step-by-step reasoning. However, most of the previous methods use heuristic or rule-based approaches to coordinate reasoning and planning, which may not be optimal. This work seeks to apply reinforcement learning to Agentic RAG and compare different approaches to better understand their capabilities and shortcomings.

II. OVERVIEW OF AGENTIC RETRIEVAL-AUGMENTED GENERATION

Agentic Retrieval-Augmented Generation (Agentic RAG) extends traditional RAG approaches by incorporating aspects of planning, adaptability, and autonomy [3], [5]. Recent advances in reasoning-centric systems and the development of chain-of-thought prompting have also contributed to this area, enabling more structured reasoning processes in language

models [9], [10]. By design, Agentic RAG approaches differ from standard retrieve-then-generate pipelines and instead employ more involved iterations and control mechanisms, thereby enabling more sophisticated reasoning than conventional methods. While there is variability in the specific design choices and implementation details, most approaches have three basic elements: the retriever, the generator, and an augmentation component. In general, the overall approach can be seen in Fig. 1, which depicts the basic three-part architecture that underlies most such systems. The three components allow the model to generate responses that take into account external information, thereby providing more contextually appropriate and informative answers.

A. Retriever Module

The retriever is a key component of most RAG systems responsible for retrieving documents from external knowledge sources relevant to the input query or intermediate representation derived from it. Traditional RAG systems perform this step only once, while Agentic RAG introduces iterative retrieval steps with contextual refinement to obtain relevant documents. This process is especially important for multi-hop reasoning, where several logical steps are needed to answer the initial question, which often requires multiple rounds of retrieval [11]. Retrieval based on graph traversal and graph-based RAG systems in general allow for more effective reasoning [12], [13]. However, the effectiveness of such systems critically depends on the accuracy of the retrieval step. A poorly retrieved document may contain misleading information that will subsequently affect the answer generation stage, leading to hallucination.

B. Generator Module

Generator The generator is usually implemented by a large language model (LLM), which generates responses by combining the input query with the retrieved contextual information. In Agentic RAG, the generator extends beyond the production of final outputs; it engages in reasoning processes, enabling it to perform sub-queries, decompose tasks, and generate step-by-step reasoning chains under the direction of planning algorithms. This improves upon the coherence, interpretability, and reasoning ability of the model [6], [9]. Self-reflective retrieval-generation systems such as Self-RAG also improve the factual consistency of generated outputs by allowing the model to self-criticize and edit its own outputs [14]. The combination of retrieved knowledge with parametric knowledge in the generator also enables more factual and contextual outputs.

C. Augmentation Module (Knowledge Base)

The augmentation comprises external knowledge sources that provide the context for retrieval. These can take many forms, such as a collection of unstructured documents, well-structured databases, knowledge graphs, domain-specific repositories, or a combination of these storage mediums. The knowledge base offers grounding and domain-specific information, allowing it to store up-to-date and specialized

information not present in the default knowledge of an large language model. The creation of efficient indexes, the implementation of semantic search, and the scalability of storage are important considerations. The diversity and quality of the knowledge base strongly impact the effectiveness of the overall Agentic RAG, as reflected by the iterative retrieval and reasoning loop in the Fig. 2.

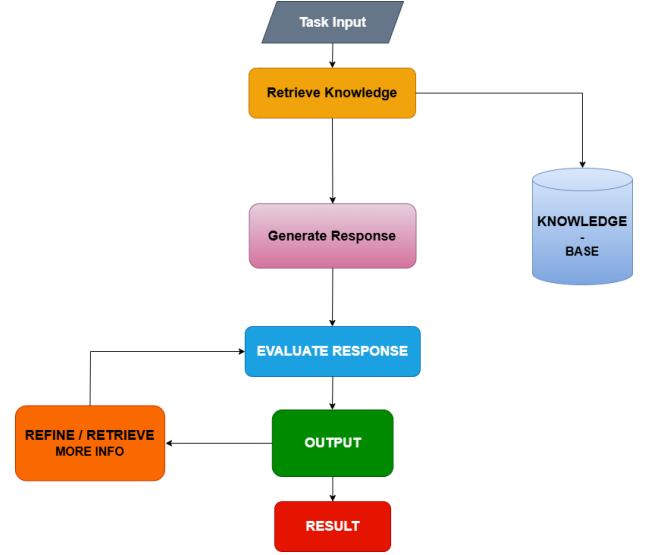


Fig. 2: Work Flow of Agentic-RAG

III. CONTRIBUTIONS

This work makes the following key contributions:

- A structured comparative analysis of existing Agentic RAG methodologies, identifying key strengths, limitations, and research gaps.
- A detailed evaluation of architectural trade-offs across multiple paradigms, including graph-based, multi-agent, and adaptive systems.
- The proposal of a Reinforcement Learning-enhanced Agentic RAG framework for adaptive retrieval and reasoning optimization.
- Identification of open challenges and future research directions for scalable and autonomous retrieval systems.

A. Comparative Analysis of Agentic RAG Methodologies

This section provides a comparison of major Agentic RAG approaches, with an emphasis on the design choices, advantages, and limitations. A comparison of these approaches in terms of datasets, advantages, and limitations is provided in Table I. The agentic RAG paradigm encompasses multiple approaches, rather than one monolithic solution, with each addressing different aspects of the original RAG paradigm. Moreover, another comparison is provided in Table II that focuses on the differences in retrieval methods, efficiency, reasoning, and scalability.

References	Methodology	Datasets	Advantages	Disadvantages
[15]	Multi-Agent Agentic RAG	Hotpot QA, Multi-hop benchmark, Multi-domain QA datasets	High accuracy, Parallel execution, Strong performance	High system complexity, debugging difficulty, expensive
[16]	Goal-Oriented Agentic RAG	FinTech enterprise knowledge base (1,624 docs), enterprise documentation, human-curated procedural QA	Strong multi-step procedural reasoning, better handling of ambiguous and complex queries, reduced hallucination	High latency, high computational and infrastructure cost, high system complexity
[17]	Traditional Agentic RAG	Natural Questions (NQ), Wikipedia-based QA datasets, enterprise document repositories, open-domain QA benchmarks	Low latency, fast response rate, simple architecture and debugging, cost efficient	No multi-step reasoning, limited hallucination control
[18]	Graph-Based Agentic RAG	Knowledge-graph QA datasets, biomedical knowledge graphs, healthcare knowledge graphs, relational multi-hop QA datasets	Graph traversal for structured reasoning, graph paths provide reasoning traces, reduced hallucination	High graph construction and maintenance cost, scalability issues, limited flexibility
[19]	Hierarchical RAG	Enterprise knowledge bases, FinTech corpora, smart city IoT datasets, OpenAI Gym simulation tasks	Efficient handling of complex tasks, improved coordination, better resource allocation	Increased latency, error propagation, complex system design
[20]	Capability-Centric Agentic RAG (Conceptual)	Tool-use benchmarks, QA corpora (general references)	Clear conceptual framework, holistic intelligence modelling	Conceptual framework only, difficult capability measurement, implementation ambiguity
[21]	Adaptive Agentic RAG	PubMed corpus, arXiv corpus, IEEE Xplore documents, Google Scholar documents, user interaction dataset, query logs and feedback data	Continuous performance improvement, better personalization	Increased computational cost, longer response time, may adapt too closely to frequent patterns
[22]	Corrective Agentic RAG	HotpotQA, scientific QA corpora, open-domain factual QA datasets, enterprise QA collections	Reduced hallucination, improved output reliability, better handling of retrieval failure	Increased latency, high cost, self-correction not always reliable
[23]	Domain-Specific Agentic RAG	Clinical guidelines, legal case law repositories, financial filings, regulatory databases, FinTech enterprise documents	High domain accuracy, curated domain knowledge, reduced hallucination	Limited generalization, high data curation cost, integration complexity
[24]	Unified-domain Agent Stack	Domain-specific LLM training (BioBERT, LegalBERT, FinBERT), regulatory databases, clinical and financial repositories, enterprise knowledge bases	End-to-end integrated intelligence, high reliability for high-stakes domains	Extremely high system complexity, high implementation cost

TABLE I: Comparative Analysis of Agentic-RAG Methodologies

B. Traditional Agentic RAG

Traditional Retrieval-Augmented Generation approaches focus on the most minimal viable system: simplicity, speed, and low cost in terms of resources. While effective in their domain, these systems sacrifice flexibility and reasoning power: iterative retrieval, task decomposition and re-planning, and adaptive retrieval strategies are all out of scope for a classic RAG system. As a result, they are ill-suited for complex multi-step reasoning tasks or ambiguous queries, and typically have a worse hallucination rate.

C. Graph-Based Agentic RAG

Graph-based RAG leverages the power of graphs and their traversals to achieve structured reasoning establishing connections between entities [12], [13]. Such methods allow for enhanced explainability and improved multi-hop reasoning by virtue of the underlying graphical depictions of connections

and relationships. These techniques also facilitate more factual grounding of response generation processes while enabling explanation pathways. However, the graph-based approaches are associated with extremely high overheads in construction and maintenance, and they also require extremely large scales, and last but not least, they are highly dependent on the accuracy of the graph quality and suffer from the incapacity to adapt to dynamically changing domains.

D. Multi-Agent Agentic RAG

Multi-agent architectures involve partitioning the task among several agents. By doing so, they allow for greater collaboration and distributed reasoning, which are beneficial for long-context scenarios and complex reasoning. At the same time, such architectures are harder to debug and reason about, require more computational power, and introduce overhead coordination and communication between agents. Addition-

Methodology	Limitations/Challenges
Multi-Agent Agentic RAG	Multi-agent coordination and communication, response conflicts across agents
Goal-Oriented Agentic RAG	Scalability issues, lack of standardized evaluation metrics, risk of API errors and tool misuse
Traditional Agentic RAG	Poor performance for complex tasks, no adaptability or learning capability, limited knowledge base
Graph-Based Agentic RAG	Dependency on graph quality, integration complexity, difficulty handling newly emerging information
Hierarchical RAG	Scalability issues due to multiple layers, lack of standard evaluation metrics, high computational cost
Capability-Centric Agentic RAG (Conceptual)	Lack of standardized benchmarks, high integration complexity
Adaptive Agentic RAG	Data dependency for learning, stability and drift issues, evaluation complexity
Corrective Agentic RAG	Risk of over-correction, benchmark gaps, lack of universal correctness metrics
Domain-Specific Agentic RAG	Data availability constraints, knowledge drift
Unified-domain Agent Stack	Scalability issues, lack of standardized evaluation frameworks

TABLE II: Identified Limitations Across Agentic-RAG Methodologies

ally, there is a risk that agents will produce contradicting outputs, resulting in faulty reasoning. Finally, multi-agent models are harder to scale up due to the increased complexity of interaction between agents.

E. Goal-Oriented Agentic RAG

Goal-driven approaches emphasize on decomposition of a task into steps and procedure-based reasoning. By incorporating a goal, these approaches excel at multi-step reasoning, mitigate against hallucination, and are better at handling complex queries in comparison to generic approaches. The problem with such methods is that they add latency and overhead since they require additional planning steps in addition to the execution steps. Furthermore, there is currently a lack of standardized evaluation methodology and the danger of misuse is high.

F. Hierarchical Agentic RAG

Organizing an activity into a hierarchy allows for control at various levels and facilitates coordination and reasoning at higher levels. At the same time, however, such systems involve increased delays due to the sequential nature of operations and are more susceptible to errors that may arise at different stages. The complexity of the system adds to its overhead and decreases its effectiveness in real-time operations.

G. Adaptive Agentic RAG

Adaptive agentic RAG systems utilize feedback-driven learning mechanisms to adapt the retrieval process and improve performance [14]. Retrieval optimization can be further supported by reinforcement learning-based approaches

enabling adaptive planning and evidence selection [25], [26]. Such methods are typically more flexible and personalized. At the same time, they have greater computational overhead and response time, require high-performance feedback data, and are challenging to evaluate due to system dynamics and potential overfitting to recurring queries.

H. Domain-Specific and Unified Agentic Architectures

The Agentic RAG Concept and its variations essay Domain-specific agentic RAG models make use of knowledge bases to gain verifiable accuracy and avoid hallucination in a specific domain, such as healthcare, legal, or finance field. Unified-domain models allow to combine domain-adapted language models with retrieval and agentic parts for end-to-end performance. However, the approach has limited applicability beyond the target domain and requires extensive data curation and maintenance. Moreover, knowledge drift, complexity, and the absence of standardized benchmarks impede broader adoption of the architecture.

I. Identified Research Gaps

The identified limitations and research gaps in various Agentic RAG paradigms serve to highlight a number of salient issues in contemporary Agentic RAG systems.

The key insights from the comparative analysis demonstrate that Agentic RAG methods currently face a number of critical research challenges. Firstly, most Agentic RAG systems exhibit significant overhead in terms of computational complexity due to the iterative retrieval, multi-agent coordination, and complex reasoning processes involved. This is especially true for multi-agent Agentic RAG systems, hierarchical architectures, and graph-based approaches, where the coordination overhead is particularly taxing on the system. Secondly, Agentic RAG methods are often challenging to scale in terms of agents, iterative cycles, and overall complexity without incurring substantial performance losses. This scalability limitation is primarily attributed to the increased overhead in agent coordination, retrieval complexity that grows exponentially with depth, and additional computational costs inherent to the reasoning process. Thirdly, it is evident that the field of Agentic RAG lacks standardized benchmarking suites to evaluate and compare the efficacy of different approaches on a common basis. The current state-of-the-art methods are evaluated using a variety of performance metrics, which complicates the ability to make direct comparisons between models. Moreover, each approach tends to be tested on different problem sets, thus rendering the assessment of their relative strengths and weaknesses in specific scenarios challenging. Fourthly, most Agentic RAG methods are based on heuristics or require extensive rule engineering. While such strategies can be effective in certain contexts, they are rarely adaptive or flexible enough to accommodate dynamically varying queries, knowledge bases, and domain contexts. With regard to the fifth issue, it is noteworthy that current Agentic RAG methods encounter unique difficulties when applied to specific domain settings. While domain-specific knowledge bases often allow for higher

reliability, they are also constrained by the limitations of the knowledge cutoff date and require additional maintenance. Additionally, domain-specific approaches often demonstrate reduced efficacy when generalized to other domains. These limitations provide further motivation to explore alternative strategies that can dynamically adapt to both queries and their context. It is evident from the preceding discussion that Agentic RAG methods would benefit from incorporating additional learning components that facilitate adaptive retrieval coordination. In particular, the application of Reinforcement Learning (RL) seems particularly promising, as it enables the optimization of retrieval policies, thus allowing agents to make optimal decisions in terms of trade-offs between accuracy and computational overhead. By reviewing the key insights from the comparative analysis, this article aims to provide structure to the discussion of Agentic RAG methods. The next section will demonstrate how Reinforcement Learning can be leveraged to build multi-level policy optimization frameworks that enable autonomous, adaptive, and efficient retrieval. In particular, the following section will discuss a novel method that leverages reinforcement learning to address the aforementioned limitations and allow agents to adapt their retrieval strategies based on observed data.

J. Reinforcement Learning-Enhanced Agentic RAG Framework

To address the shortcomings of the existing Agentic RAG methods, we propose a novel Reinforcement Learning (RL)-based Agentic RAG framework. By incorporating learning-based optimization mechanisms, our approach enables enhanced adaptability, improved decision-making processes, and efficient multi-step reasoning within the retrieval and reasoning paradigm. Furthermore, unlike the previously described methods that mostly utilize rule-based algorithms for planning and retrieval, the proposed solution employs a novel learning-based mechanism for optimizing both planning and retrieval. Fig. 3 provides an overview of our RL-based Agentic RAG framework, which decomposes the high-level planning and low-level retrieval policies. By utilizing hierarchical reinforcement learning, the authors are able to decouple high-level planning from low-level execution [26]. Most recently, several works [25], [27], [28] demonstrated that reinforcement learning could be leveraged to optimize graph retrieval, search planning, and reasoning. It accomplishes that by introducing a hybrid of high-level and low-level policies. In turn, such an approach provides increased flexibility and adaptability while balancing global planning and immediate response decisions, ultimately facilitating more effective multi-step reasoning processes.

K. Working Mechanism

The proposed framework is organized around a two-layered policy:

1) *High-Level Policy: Goal Structuring Layer:* The high-level policy breaks down the user query into sub-goals and determines the order of reasoning to achieve them. It plans out a sequence of reasoning steps toward the goal by engaging

in goal-oriented reasoning. The high-level policy is optimized with long-term rewards, such as answer correctness, reasoning consistency, and hallucination penalties. By learning over extended decision horizons, high-level policies enable improved planning abilities and thus help the model with multi-step reasoning.

2) *Low-Level Policy: Adaptive Retrieval Layer:* The low-level policy handles all retrieval-related decisions, such as query rewriting, document selection, tool calling, and iterative refinement of retrieved evidence. The low-level policy makes decisions on how to best perform retrieval based on intermediate feedback and contextual information. Using the combination of a high-level policy and a low-level policy allows the system to operate at different levels of abstraction while still being able to effectively plan and execute the reasoning process. The high-level policy provides guidance and oversight, while the low-level policy is more focused on executing specific tasks and adapting to changing circumstances.

3) *Policy Update and Reward Mechanism:* Both policies are updated in an iterative manner using reinforcement learning. The environment gives rewards according to several evaluation criteria such as

- Factual response
- Relevant and quality documents retrieved
- Reasoning consistency and coherence
- Speed of retrieval
- Resource costs

These reward signals guide the learning process to maximize the success rate of tasks and minimize hallucinations as well as unnecessary retrieval and reasoning steps. With such rewards, the policy learned by an agent can perform retrieval and reasoning appropriately for different types of queries and domains. An illustration of the interactions among policies and the environment is shown in Fig. 4.

L. Potential Impact and Advantages

The incorporation of reinforcement learning into Agentic RAG offers several advantages, such as :

1) *Improved Retrieval and Reasoning Accuracy:* the ability to learn which retrieval strategies produce the most informative and accurate results via feedback-driven optimization [25], while reasoning-enhanced retrieval and reinforcement learning methods can facilitate adaptive search behaviors and improve factual reasoning [10], [26] to reduce hallucination and improve confidence in the reasoning process.

Example: In scientific research scenarios, the system can prioritize retrieval from high-quality sources, improving the accuracy and reliability of the information.

2) *Adaptive Multi-Step Task Execution:* The hierarchical model allows for breaking down the query into sub-problems and iteratively refining the results based on intermediate answers, facilitating complex multi-hop reasoning.

Example: In the context of medical information search, the query can be decomposed into the analysis of symptoms, literature search, and treatment recommendation.

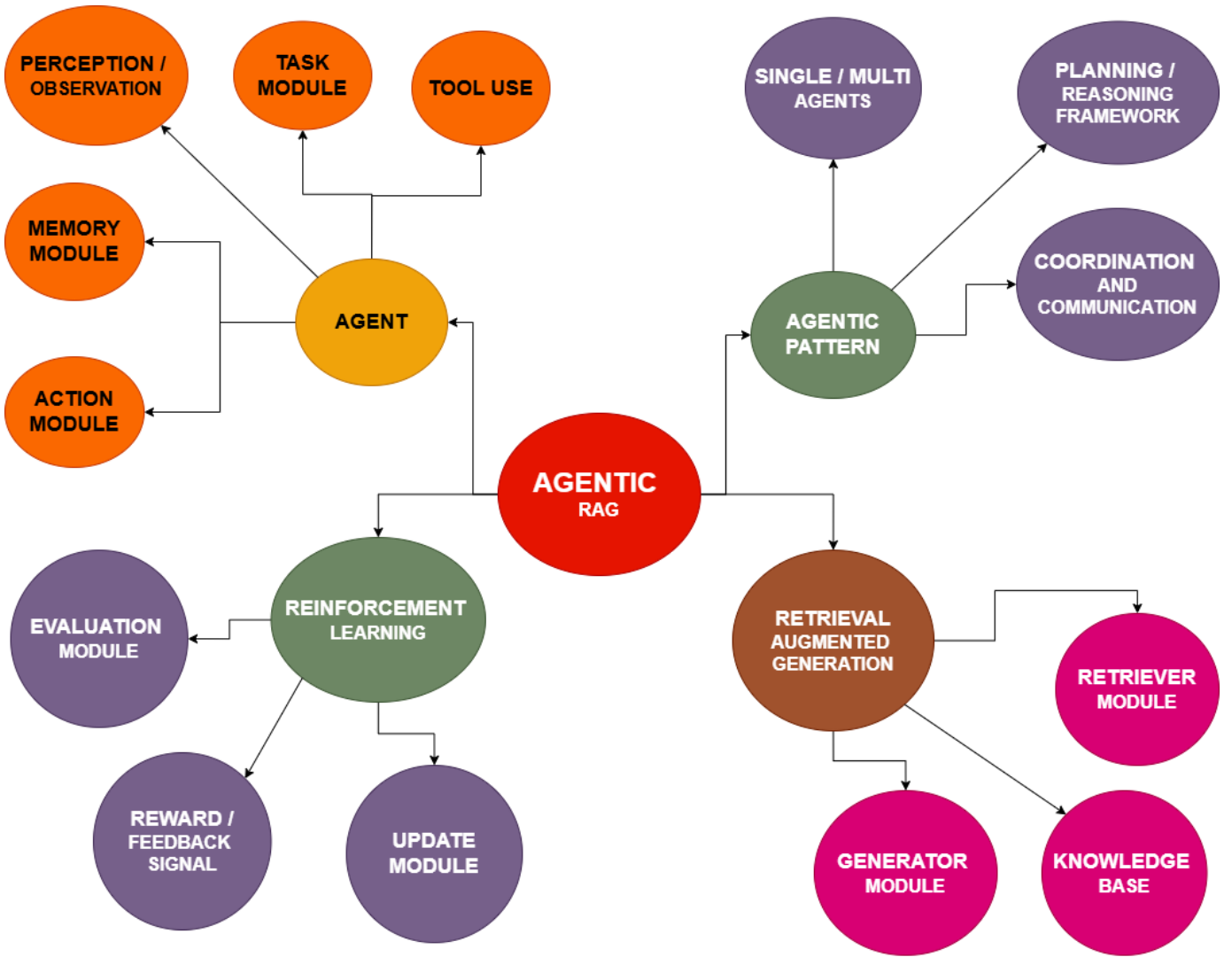


Fig. 3: Agentic-RAG + Reinforcement Learning

3) *Scalability and Efficiency Optimization:* Reinforcement learning enables the system to learn the best possible ways, in this case of database searching, thereby saving computation time on nonoptimal algorithms. This can result in significant gains in search speed, compared to a system using fixed rules for searching.

4) *Enhanced Autonomy and Continuous Learning:* The framework is designed to continuously refine itself through feedback mechanisms, allowing it to operate independently and adapt to shifting data distributions, evolving domains, and changing user preferences with minimal human intervention.

M. Challenges and Limitations

Even though it has its benefits, the proposed RL-driven Agentic RAG framework has some of its shortcomings:

1) *High Training Complexity:* Training the hierarchical reinforcement learning models requires much data and is highly dependent on the reward function design, which can be error prone.

2) *Computational Overhead:* The combination of RL with multi-step retrieval and reasoning increases computational cost for both training and inference, which may hinder its applicability in real-time scenarios.

3) *Exploration vs. Exploitation Trade-off:* Balancing the exploration of new retrieval strategies and the exploitation of the optimal one is a challenging task since both insufficient exploration and overuse of a certain strategy lead to sub-optimality.

4) *Evaluation Complexity:* Evaluating RL-based Agentic RAG systems is not a trivial task, as their nature of operation is dynamic and the interactions that take place are long-term. The standard benchmarks often fail to account for policies' effectiveness and learning progression.

5) *Data Dependency and Reward Bias:* The performance of the system is largely determined by the quality of the feedback signals. In the case of learning from rewards, rewarding policy may produce incorrect learning signals, thus leading to poor overall performance and generalization ability.

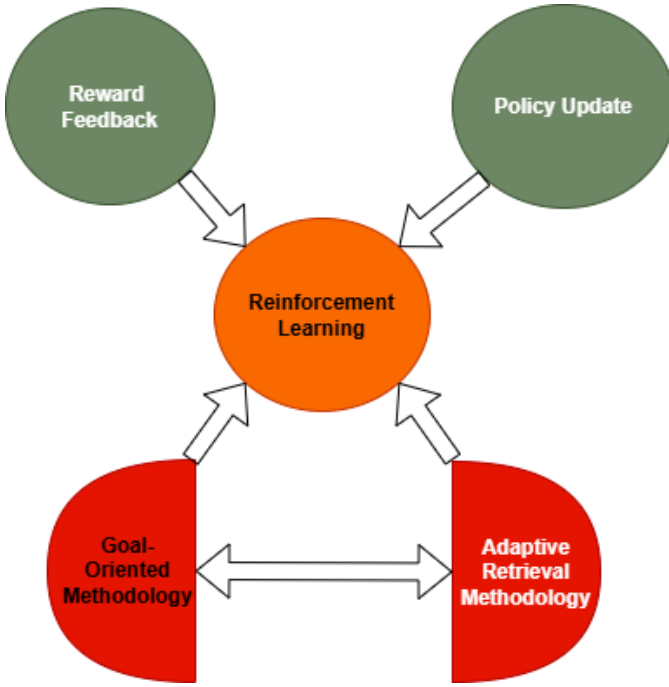


Fig. 4: Reinforcement Learning Framework

6) *System Stability and Convergence*: Finally, establishing stable convergence properties is non-trivial when dealing with hierarchical policies, as interactions between higher-level and lower-level policies can cause instability.

N. Discussion

The proposed RL-enhanced Agentic RAG framework indeed addresses several of the aforementioned limitations, notably lack of optimization and scalability. By incorporating learning-driven coordination mechanisms, the framework manages to reconcile reasoning accuracy, adaptability, and computational efficiency. While training complexity and assessment present significant challenges, the integration of reinforcement learning mechanisms appears to be a promising direction in the development of next-generation retrieval augmented models [26], [27]. The emergence of reasoning-centric language models and the increasing need for adaptive retrieval frameworks also suggest the potential value of autonomous and learning-driven retrieval approaches. Subsequent research endeavors could prioritize the optimization of reward functions, reduction of computational demands, and the establishment of standardized evaluation benchmarks for RL-enhanced Agentic RAG solutions.

IV. CONCLUSION AND FUTURE DIRECTIONS

This work presented a comparison of Agentic Retrieval-Augmented Generation approaches and investigated the application of reinforcement learning methods to address their identified shortcomings. While Agentic RAG methods indeed facilitate more competent reasoning and fact-grounded responses, the surveyed approaches are often too simplistic

and restrained in their ability to adapt. The authors proposed an alternative approach based on reinforcement learning principles that allows for more flexible retrieval and structured decision-making processes. However, this solution comes at the cost of increased computational complexity and the necessity of additional training and assessment procedures. Further research should therefore prioritize the reduction of computational costs, the improvement of reward functions, and the formulation of standardized evaluation criteria. Apart from these aspects, there is a need to increase the transparency of the decision-making process and consider the possibility of implementing human feedback to obtain more reliable results. Overall, the application of reinforcement learning methods to Agentic RAG models is a promising approach to improve the abilities of question-answering systems [5], [25], [26].

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